

Mohammad Reza Kianifar

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Skills

Programming Languages Python, Java, Kotlin, JavaScript, TypeScript, Bash, PHP, C++

Web Django, React, Spring Boot, Laravel, RPC, GraphQL, REST

Big Data & Databases PostgreSQL, MySQL, MongoDB, Kafka, Celery, Redis, Cassandra, Elastic Search

Site Reliability Engineering Docker, Kubernetes, GKE, AWS, Jenkins, GitHub Actions, Datadog, Circle CI

ML Engineering LLM Evaluation, Agentic Systems, Benchmarking, LLM-as-a-Judge, Prompt Engineering

Software Engineering & Tools Microservices, Object-oriented patterns, Agile (Scrum), Git, Unit testing

Education

Master of Science in Computer Science University of Calgary, Canada; GPA 3.93/4; CCPC 4th place (2022, 2023) Jan 2020 - Sep 2021

Bachelor of Science in Software Engineering University of Tehran, Iran; GPA 3.79/4 (top 20%) Sep 2015 - Feb 2020

Professional Experience

Senior ML Software Engineer Turing, Canada Oct 2025 - Present

- › Built production platform on the [Harbor](#) framework for authoring, evaluating, and reviewing agentic tasks (e.g., Terminal-Bench, SWE-Bench) with coding agents including Terminus-2, Claude Code, and Codex.
- › Scaled the platform to 500+ monthly active users across 22 teams, running 300K+ evaluations and 130K+ verifications per month over 100K+ tasks.
- › Designed a **Celery**-based concurrency scheduler with per-user limits and PostgreSQL advisory locks, running 1000+ concurrent agentic trials without deadlocks.
- › Developed chat-based **agentic verification** with dynamic context selection that reviews tasks and automatically fixes issues.
- › Built **LLM-driven task generation** that extracts candidates from code repositories and produces benchmark-specific Harbor tasks across harness levels, generating 1.2K+ tasks from 400+ repositories.
- › Created a **pipeline system** with a visual builder that lets teams design, reuse, and automate task review, executing 35K+ pipeline runs within a month of launch.
- › Designed and implemented the harness for a data-science benchmark that connects agents to Jupyter notebook environments via MCP, with rubric-based LLM-as-a-judge grading.
- › Implemented ledger-based credit accounting that reserves and settles LLM, training, and sandbox costs per user and team.
- › Tech stack: **Python, Django, PostgreSQL, Celery, Redis, GKE, Google Cloud Storage, TypeScript**, and **React**

Google Summer of Code Contributor Zulip, Remote May 2025 - Sep 2025

- › Migrated 1:1 and self direct messages in [Zulip](#) (26K+ GitHub stars, 1,500+ contributors, used by thousands of organizations) from the legacy **Personal** recipient model to a unified **DirectMessageGroup** model to enable per-conversation features such as pinning and hiding DMs.
- › Merged [18 PRs](#) in a phased, backward-compatible rollout covering message sending, unread counts, email and push notifications, scheduled messages, analytics, and Slack, Mattermost, and Rocket.Chat imports.
- › Wrote a batched backfill migration that moves existing records to the new model with minimal downtime, using **Python, Django**, and **PostgreSQL**.

Software Engineer *Wave HQ, Canada*

Jan 2024 - Sep 2025

- > Contributed to the risk management systems at [Wave](#), a financial platform with 200K+ active small business users, using **Python, Django, PostgreSQL, RPC, GraphQL** and **React**.
- > Led the architecture design and end-to-end implementation of Apple Pay integration.
- > Developed an automated schema diagram generator for APIs and Kafka events used across teams.
- > Established clean code and review principles to improve collaboration and code quality across a 13-person engineering team.

Graduate Research Assistant *University of Calgary, Canada*

Jan 2021 - Jan 2024

- > Published a peer-reviewed journal article in [MDPI](#) on *Graph Generalization for Software Engineering*.
- > Designed an algorithm that compares directed labeled graphs, abstracting differences into variable components, with 90% test coverage.
- > Detected important semantic changes in library version updates with 54% higher precision than alternative tools.

Lead Software Engineer, Software Engineer [Tapsell, Iran](#) (70M+ monthly active users) Jan 2019 - Dec 2020

- > Led a cross-functional team of 12 engineers and designed a video-ad integration with [a partner company](#) handling 3M+ daily requests.
- > Built a data collector handling 5M+ daily requests with **Kafka, Cassandra**, and **Elasticsearch**, and helped replace a Spark-based service with stream processing.
- > Contributed to 12 microservices in **Java, Kotlin**, and **Spring Boot**, deployed with **Jenkins** and **Kubernetes**.
- > Improved dashboard API performance and resource use by 55% by optimizing **PostgreSQL** and **MongoDB** queries, and removed 4,000+ lines of legacy code.

Software Engineer [Hinza Co., Iran](#)

Jan 2016 - Dec 2018

- > Built a time-tracking app with 95% test coverage using **React Native, Django**, and **MySQL**.
- > Developed e-commerce platforms, including [Kitline](#), and a CMS used on 20+ websites, using **PHP** and **Laravel**.

 Leadership**President, Computer Science Graduate Society** *University of Calgary*

Jun 2021 - Apr 2023

- > Raised \$10K+ in funding, held 10+ events, and received the \$3,000 Rizvi Family Scholarship for volunteer service.

 Honors and Awards**Winner of CCPC** *Ranked 4th for 2 years in a data structure and algorithm contest*

Apr 2022, Apr 2023

Ranking in the Bachelor's Graduating Class *among the top 20% with GPA of 3.79/4*

Feb 2019

Ranking in the Iranian Nationwide University Entrance Exam *364th among 180K+ (top 0.2%)*

Jul 2015

 Volunteer Experience**President, Vice-President Social** *Computer Science Graduate Society*

Jun 2021 - Apr 2023

- > Raised over \$10,000 of funding and held 10+ events for graduate students at UCalgary.
- > Awarded Rizvi Family Scholarship in the amount of \$3,000 for volunteer services.

Technical Mentor *Summer of Code Event at the University of Tehran*

Jul 2018 - Sep 2018

- > Coordinated a team of 5 students to develop a social network for holding events at the university, using **NodeJS** and **React**.